

|              |                                          |
|--------------|------------------------------------------|
| Table 2      |                                          |
| Ethan Kepros | William Nimtz                            |
| Sunyoung You | Sujal Vajire<br>Irene Blanco<br>Casallas |

|                 |                |
|-----------------|----------------|
| Table 3         |                |
| Max Hortop      | Lautaro Garcia |
| Dashiel Matlock | Lin Tian       |

|                     |                                                       |
|---------------------|-------------------------------------------------------|
| Table 4             |                                                       |
| Jessica Duran       | Zohreh Doroudian<br>Mohamed<br>Abdulazim Ali Afifi El |
| Aidana Bekbulatkyzy | Habet                                                 |

|                 |                 |
|-----------------|-----------------|
| Table 5         |                 |
| Sam Hu          | Jack Bajcz      |
| Basma AlMahmood | Samuel Goodluck |

|             |                 |
|-------------|-----------------|
| Table 1     |                 |
| Abraham Rai | Alexander Lover |
| Xin Dong    | Jingyi Ge       |

|                 |             |
|-----------------|-------------|
| Table 7         |             |
| Saksham Mamtani | Andrew Tran |
| Sanjida Meherin | Mi Zhou     |

|                |                   |
|----------------|-------------------|
| Table 6        |                   |
| Madison Mercer | Md Shariful Islam |
| Grace Akintan  | Michael Pittman   |

## CMSE 801 - Day 3

Calculating savings growth using lists and loops

Jan 20, 2026



## General Course Announcements

- Office hours and help room hours have started this week! **You can find the schedules times and location on the course homepage.**
- Remember: at the end of class **you will be uploading your in-class notebook to D2L.**

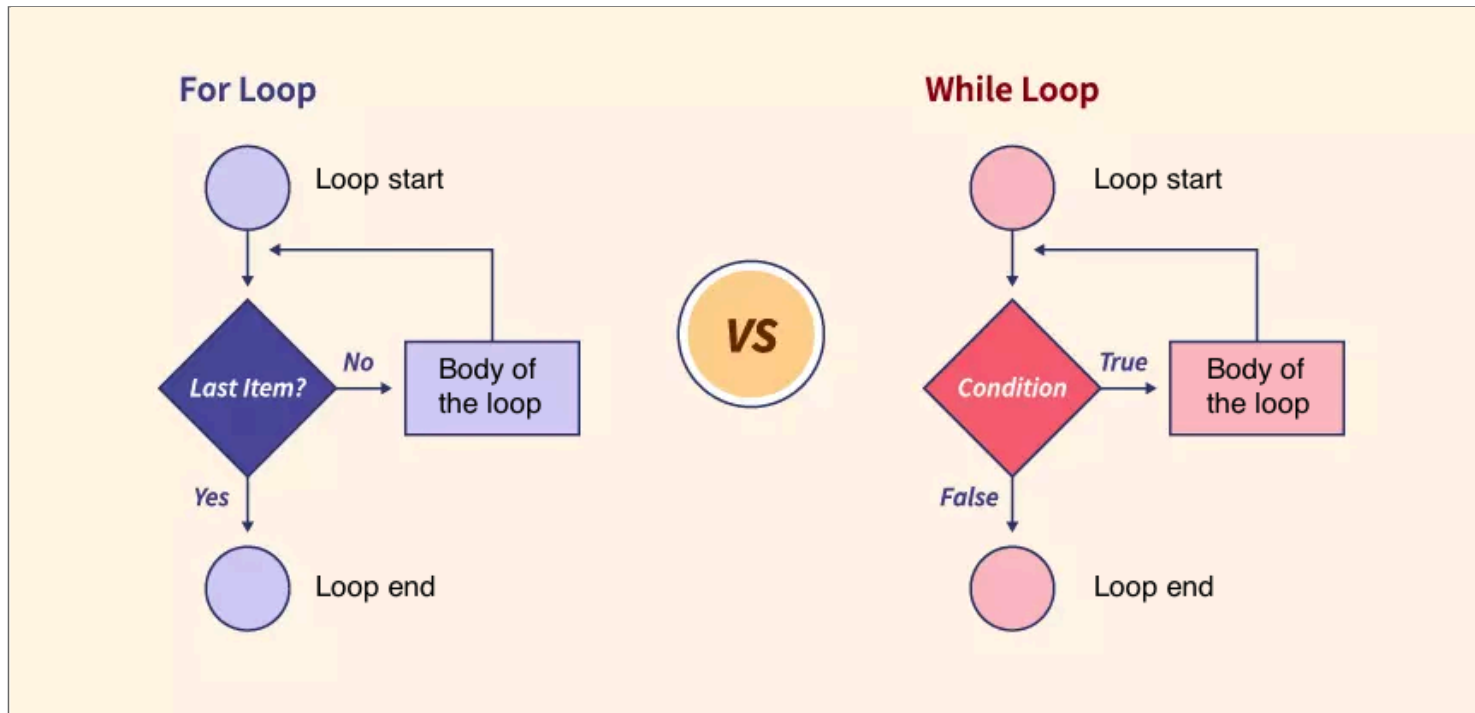
## Questions/comments from Day 3 PCA

- Some confusing terminology and jargon coming up here and there. e.g. things "initialize" and "iterate"
- Some challenges with understanding what exactly a question is asking you to do.
- When/why would I use a nested loop?
- What are the key differences between `for` loops and `while` loops?
  - **We'll spend a bit of time talking/thinking about this**
- Some struggles with Question 5 and Question 6
  - **We'll take a look at some possible solutions and some potential pitfalls**

The `for` loop vs. the `while` loop

What's the key difference?

## The for loop vs. the while loop



Let's look at some code examples

In [11]:

```
# Loop by integer values or "by index"  
for i in range(0,3):  
    print(i)
```

0  
1  
2

In [13]:

```
# Loop by value in a list  
mylist = ['red', 'yellow', 'blue']  
for value in mylist:  
    print(value)
```

red  
yellow  
blue

In [9]:

```
# Loop until the the "count" value is greater than or equal to 3  
count = 0  
while count < 3:  
    print(count)  
    count += 1
```

0  
1  
2

## Reviewing Question 5

Below is a possible solution to Question 5, will it work? If not, what's missing?

In [14]:

```
x = [2,4,6,8,10,12,14,16,18]
y = [10,8.25,7.5,7,6.5,7,7.5,8.25,10]

# Create a loop to add the x and y values together and store them in a list called "addition"
for i in range(len(x)):
    # index the x and y lists, add their values, and append them to the list
    addition.append(x[i] + y[i])

# Check to see if it worked
print(addition)
```

---

```
NameError                                Traceback (most recent call last)
Cell In[14], line 7
      4 # Create a loop to add the x and y values together and store them i
n a list called "addition"
      5 for i in range(len(x)):
      6     # index the x and y lists, add their values, and append them to
the list
----> 7     addition.append(x[i] + y[i])
      9 # Check to see if it worked
     10 print(addition)
```

NameError: name 'addition' is not defined

This should work!

In [15]:

```
x = [2,4,6,8,10,12,14,16,18]
y = [10,8.25,7.5,7,6.5,7,7.5,8.25,10]

# Initialize my empty list
addition = []

# Create a loop to add the x and y values together and store them in a list called "addition"
for i in range(len(x)):
    # index the x and y lists, add their values, and append them to the list
    addition.append(x[i] + y[i])

# Check to see if it worked
print(addition)
```

```
[12, 12.25, 13.5, 15, 16.5, 19, 21.5, 24.25, 28]
```

## Reviewing Question 6 (the nested loop!)

What if for each individual  $x$  value, we want to calculate  $(x + y)^2$  for *all* of the  $y$  values and print the list for each  $x$  value?

In [16]:

```
x = [2,4,6,8,10,12,14,16,18]
y = [10,8.25,7.5,7,6.5,7,7.5,8.25,10]

# Time to nest some loops!
for i in range(len(x)):
    xy_squared = []
    # index the x and y lists, add their values, and append them to the list
    for ii in range(len(y)):
        xy_squared.append((x[i] + y[ii])**2)

    # Check to see if it worked
    print(xy_squared)
```

[144, 105.0625, 90.25, 81, 72.25, 81, 90.25, 105.0625, 144]  
[196, 150.0625, 132.25, 121, 110.25, 121, 132.25, 150.0625, 196]  
[256, 203.0625, 182.25, 169, 156.25, 169, 182.25, 203.0625, 256]  
[324, 264.0625, 240.25, 225, 210.25, 225, 240.25, 264.0625, 324]  
[400, 333.0625, 306.25, 289, 272.25, 289, 306.25, 333.0625, 400]  
[484, 410.0625, 380.25, 361, 342.25, 361, 380.25, 410.0625, 484]  
[576, 495.0625, 462.25, 441, 420.25, 441, 462.25, 495.0625, 576]  
[676, 588.0625, 552.25, 529, 506.25, 529, 552.25, 588.0625, 676]  
[784, 689.0625, 650.25, 625, 600.25, 625, 650.25, 689.0625, 784]

## Goals for today

- Get some practice and comfort with writing loops and using lists in Python.
- Use these tools to track accumulated savings over time using a retirement investment model.

Looking forward:

Exploring conditional statements and a brief intro to functions in the Day 4 PCA

- " If " statments in Python (and their more complex " elif " and " else " statements)
  - Used to control code flow
- Functions in Python
  - Modular
  - Reusable
  - Useful!

**Note:** this might also feel like a slightly large PCA, but you have a few days to work though it, I encourage you to tackle it sooner than later.